Pre-requisites

The AI Flood Prediction System was developed using Python and several open-source libraries for data preprocessing, machine learning, visualization, and web application deployment. The following software tools and libraries are required to build and run the project successfully.

Software / Tool	Purpose
Python 3.12	Programming language used to develop the application
Anaconda Navigator	Package and environment management
PyCharm Community Edition	Python IDE for coding and debugging
NumPy	Numerical computations and array operations
Pandas	Data preprocessing and analysis
Scikit-learn	Machine learning model development
XGBoost	Gradient boosting classification algorithm
Matplotlib	Data visualization
Seaborn	Statistical data visualization
Flask	Web application framework
Joblib	Saving and loading trained models
HTML5, CSS3, Bootstrap	Front-end user interface

Software Requirements

- Operating System: Windows 10/11
- Python Version: 3.12 or above
- IDE: PyCharm Community Edition
- Environment Manager: Anaconda Navigator
- Web Framework: Flask
- Browser: Google Chrome / Microsoft Edge
- **Python Libraries**

- Install the required libraries using:
- `pip install flask pandas numpy matplotlib seaborn scikit-learn xgboost joblib`

Download Links

- Python – https://www.python.org/downloads/
- Anaconda Navigator – https://www.anaconda.com/download
- PyCharm – https://www.jetbrains.com/pycharm/
- NumPy – https://numpy.org/
- Pandas – https://pandas.pydata.org/
- Scikit-learn – https://scikit-learn.org/
- XGBoost – https://xgboost.readthedocs.io/
- Matplotlib – https://matplotlib.org/
- Seaborn – https://seaborn.pydata.org/
- Flask – https://flask.palletsprojects.com/